DynaByte: Shared-Budget Allocation of Expert Precision, Residency, and Lookahead

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On a single GPU, expert memory and host-to-device bandwidth are shared among fidelity, residency, and lookahead. A byte assigned to one use cannot retain another expert or stage a future one, and a policy that fixes two uses and tunes the third can still raise tail latency. DynaByte admits one exchange at a time. The exchange names the lease it requests and the leases released to fund it, and it is admitted only when that release lowers predicted exposed wait and $\widehat{R} \le \widehat{\epsilon}$. A representation is published only after its copy completes, and the executor holds the expert-memory peak within budget. $\widehat{R} \le \widehat{\epsilon}$ is the controller's admission estimate; it is not the measured constraint $\Delta Q \le \epsilon$. On the five downstream tasks, weighted by sample size, DynaByte's score is 1.18%, 0.64%, and 1.03% below the all-high reference on Qwen3-30B, Qwen3-Next-80B, and Phi-3.5-MoE. Under the same expert-memory budget, it reduces p95 TPOT by 11.6% relative to HOBBIT on Qwen3-30B.

CCS Concepts: • **Computing methodologies** → *Artificial intelligence*; • **Software and its engineering** → **Memory management**; • **Computer systems organization** → *Architectures.*

Additional Key Words and Phrases: mixture of experts, memory management, prefetching, quantization, performance modeling

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1 INTRODUCTION

A Mixture-of-Experts (MoE) layer routes each token to a small subset of experts, so capacity can grow without a proportional increase in per-token computation [15, 29, 38]. The expert weights still have to be stored or fetched. Models from Mixtral and Qwen3 to DeepSeek-V2 and GLaM keep tens to hundreds of billions of parameters, and in the largest case a trillion, while activating only a fraction of them per token [8, 10, 23, 48]. On one GPU those weights compete with non-expert parameters, the KV cache, activations, and runtime workspaces. When the experts cannot all remain at the desired precision, the system has to compress them and move them across the host-to-device link.

How should a limited GPU allocate its expert memory and host-to-device transfer bandwidth among fidelity, residency, and lookahead, so that Mixture-of-Experts inference reduces tail latency, keeps measured quality within ϵ, and holds the expert-memory peak within budget? These two budgets meet in three uses, defined in Section 2.1. A byte assigned to fidelity cannot also retain another expert or stage a future one, and host-to-device bandwidth decides whether that early transfer finishes before its deadline.

Prior work has effective policies for parts of this problem. Quantization and mixed-precision methods choose expert representations under a memory or accuracy target [12, 17, 31]. Expert caches and prefetchers decide which experts to keep and when to transfer missing experts, commonly under a fixed representation size [14, 39, 41, 47]. Mixed-precision offloading shows that lower-precision transfers, caching, and prefetching can run in one system [20, 43].

Section 2 measures four facts that a fixed policy does not absorb. Perplexity rises as more experts move to the lower tier, and the higher tier is the larger copy. WikiText, GSM8K, and HumanEval dispatch different experts at the same layer. Prefill and decode push the traffic that lookahead cannot

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hide in opposite directions. Raising the high-precision share changes both how many experts can stay resident and which staging depth hides their transfers.

These measurements are one allocation. Fidelity, residency, and lookahead spend the same expert-memory bytes and the same host-to-device queue. Section 3 specifies the controller that allocates them.

We make three contributions.

- We observe that expert memory and host-to-device bandwidth form one budget with three uses. A byte spent on fidelity cannot retain another expert or stage a future one, and a transfer that lookahead starts still has to finish before it is used (Section 2). A fixed partition or a fixed priority cannot compare these costs as the routed experts and the prefill–decode mix change, and systems that already combine quantization, caching, and prefetching still admit the three uses separately.
- We design DynaByte, which allocates this budget by admitting one exchange at a time. The executor holds every transition's expert-memory peak within a hard budget B. The exchange names the lease it requests and the leases released to fund it, and the controller admits it only when that release lowers predicted exposed wait and $\widehat{R} \le \widehat{\epsilon}$. That quality test estimates $\Delta Q \le \epsilon$; it is not a measured-quality guarantee. Fidelity, residency, and lookahead share that path and one byte account, and a representation is published only after its copy completes.
- We evaluate DynaByte on one GPU with Qwen3-30B, Qwen3-Next-80B, and Phi-3.5-MoE. Latency replays a fixed order of MMLU-Pro, GPQA, AIME25, GSM8K, and HumanEval, split by input/output length, at batch size 8. Quality uses WikiText-2 together with the same five tasks, weighted by sample size (200, 198, 30, 200, and 164). DynaByte's score is 1.18%, 0.64%, and 1.03% below the all-high reference on Qwen3-30B, Qwen3-Next-80B, and Phi-3.5-MoE (Table 4). Under the same expert-memory budget, p95 TPOT is 11.6% lower than HOBBIT on Qwen3-30B, 23% lower on Qwen3-Next-80B, and 27.1% lower on Phi-3.5-MoE (Table 5).

2 BACKGROUND AND MOTIVATION

2.1 Expert Memory on One Accelerator

An MoE layer routes each token to k_ℓ experts. The router determines which experts execute. Expert-choice routing lets each expert select its tokens [58]. It does not determine where their weights reside or when they arrive on the device. A selected expert that is absent must be transferred from host memory, and a demand transfer exposes that time on the forward path.

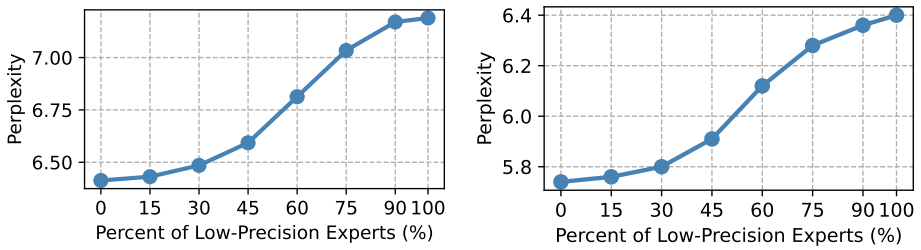
Memory for non-expert parameters, the KV cache, activations, and fixed workspaces is reserved first. The remaining capacity is the expert-memory budget B. Host memory stores a packed low-precision representation and a packed high-precision representation of every routed expert, with aggregate footprints P_{lo} and P_{hi}. We write $\Phi = B/P_{\text{lo}}$ and $\rho = P_{\text{hi}}/P_{\text{lo}}$. At any instant an expert is absent or resident at low or high precision. The forward pass can use only a published resident representation.

The host-to-device bandwidth budget is the measured copy rate C of the migration queue. C is the rate of that one queue. Representation size, whether the expert stays resident, and when its transfer occupies the queue decide which transfers consume this rate.

The same expert-memory bytes have three uses. The extra bytes of a higher-precision representation are *fidelity*. *Residency* keeps an expert on the device so that a later access requires no transfer. *Lookahead* stages an absent expert early enough for its transfer to overlap preceding computation. Lookahead is a timed transition from absence to residency. It is not a separate representation. Table 1 is the name used for each of these objects in the rest of the paper.

Table 1. Names used after this section. Precision denotes a stored tier, low or high.

Term	Meaning
Fidelity	Extra bytes of the high-precision representation, $\delta m_i^{\text{fid}} = m_i^{\text{hi}} - m_i^{\text{lo}}$.
Lease	$z = (u, i, b, [t_s, t_f))$, with use $u \in \{\text{fid}, \text{res}, \text{look}\}$.
Donor	A lease released to fund another lease.
Closure	Attaching a donor set D to a lease, forming the exchange $x = (Z^+, D, \mathcal{T})$.
Envelope S_t	Maximum routed-layer depth for candidate generation. It reserves no memory or bandwidth.
ϵ, $\widehat{\epsilon}$	ϵ bounds measured ΔPPL. $\widehat{\epsilon}$ bounds the admission quantity $\widehat{R}$.



(a) Qwen3-30B: FP16/INT4 (b) Qwen3-Next-80B: INT4/INT2

Fig. 1. WikiText-2 perplexity under frozen precision-quota sweeps. Experts are demoted in the order defined by an independent, fixed ranking; these are not online-policy results.

2.2 Fidelity Has a Quality Price and a Transfer Cost

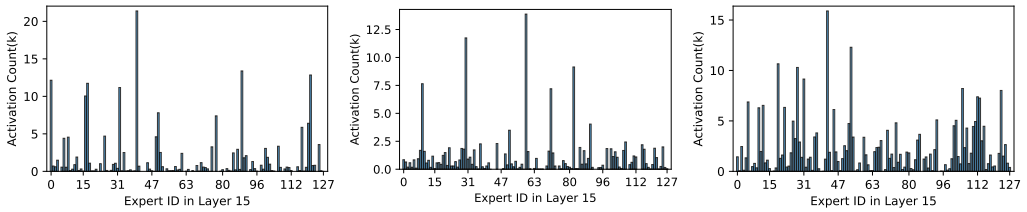
Figure 1 changes the fraction of experts stored at low precision. Each point demotes a longer suffix of one fixed ranking, so the curve prices a fidelity quota. The ranking stays frozen, and the sweep does not reassign which experts receive the higher-precision bytes.

WikiText-2 perplexity rises on both Qwen3-30B (FP16/INT4) and Qwen3-Next-80B (INT4/INT2) as more experts move to the lower tier. The rise is continuous, so a quality-degradation limit selects how large the low-precision fraction may become. The higher-precision representation is also the larger packed copy. Raising the fidelity share therefore spends expert memory and increases the bytes a later transfer must move. Which experts should receive those bytes is still open.

2.3 The Experts a Workload Uses Also Change

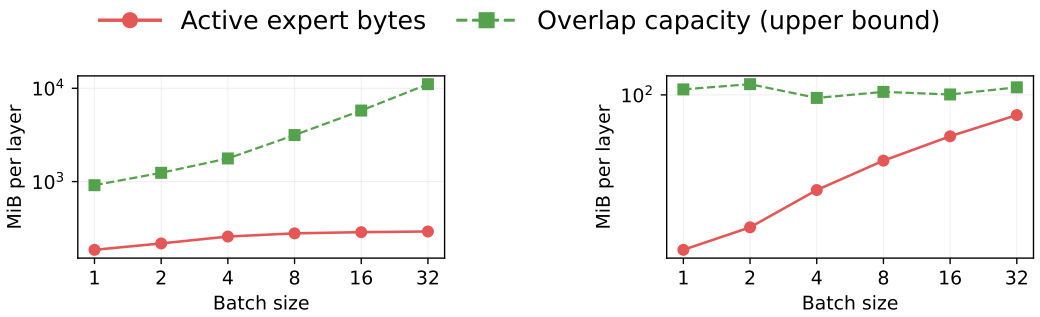
The quality curve leaves the recipients unnamed. Figure 2 counts the token–expert dispatches at Layer 15 of Qwen3-30B on WikiText, GSM8K, and HumanEval. Every expert stays at low precision, and the counters reset between workloads, so the histograms show which experts each workload uses.

The three tasks dispatch different experts. A fidelity assignment or a resident set taken from one workload therefore misses experts the next workload uses. The counts record access frequency. They leave quantization sensitivity unmeasured, so they do not rank the quality harm of a low-precision expert. Both the fidelity recipients and the resident set are workload-dependent inputs to the same allocation.



(a) WikiText (b) GSM8K (c) HumanEval

Fig. 2. Selected token–expert dispatch counts at Layer 15 of Qwen3-30B. All experts remain at low precision and the counters are reset between workloads.



(a) Prefill (b) Decode

Fig. 3. Active expert bytes and an upper bound on overlap capacity for Qwen3-30B. The bound uses measured blocking-copy bandwidth and mean per-layer TTFT or TPOT; it is not the result of an overlapping offload runtime.

2.4 Whether the Remaining Bytes Can Be Hidden Also Changes

The quality curve prices the fidelity quota, and the dispatch counts show that the accessed experts change with the task. What remains open is whether an expert that is still absent can arrive before it is used. Figure 3 compares, for Qwen3-30B, the bytes selected in one MoE layer with an upper bound on the bytes that can move during that layer's compute.

Demand comes from five measured prompt blocks at each batch size. The bound multiplies measured blocking host-to-device bandwidth by mean TTFT or TPOT and divides by 48 routed layers. The bound is an optimistic ceiling: it ignores each expert's deadline and the bandwidth concurrent kernels take. It is the capacity of the measured copy rate, not the result of an overlapping offload runtime.

Prefill touches most routed experts, and its compute window grows with the batch, so the overlap bound rises with demand. Decode touches fewer experts, but its shorter window leaves less time before use. The gap between selected bytes and this bound is the traffic lookahead cannot hide at the measured rate. Prefill and decode move that gap in opposite directions, so one lookahead depth does not fit both phases. Representation size stays fixed in this comparison.

2.5 A Fidelity Change Moves Residency and Lookahead

Figure 4 places fidelity, residency, and lookahead on one expert-memory budget at the measured host-to-device bandwidth of the replay. The experiment replays a measured 2,048-token routing

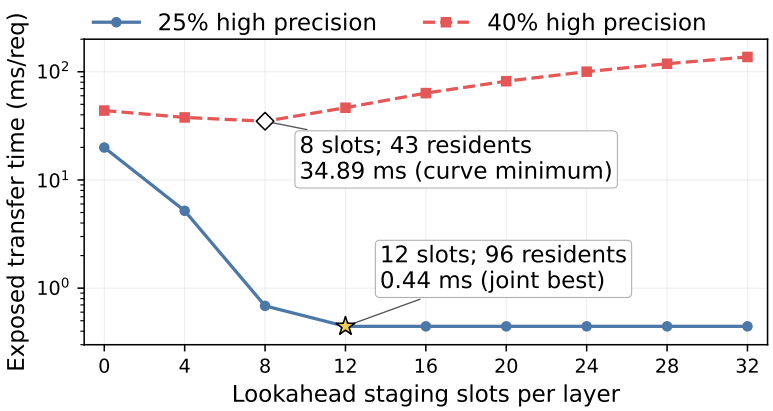


Fig. 4. Trace replay of the two quality- and memory-feasible fidelity allocations on Qwen3-30B, at the measured host-to-device bandwidth of this trace. Increasing the high-precision share from 25% to 40% reduces residency and moves the best staging allocation from 12 to 8 slots. Bandwidth is not swept. Lookahead uses oracle knowledge of the measured active set; this is not an end-to-end DynaByte result.

trace under the byte-accurate expert budget used by the Qwen3-30B runtime. An illustrative WikiText-2 limit of 2.1 perplexity above the all-high configuration leaves only the 25% and 40% high-precision fractions feasible among the scanned shares.

Residents follow the frequency ranking from warm-up traces. Lookahead has oracle knowledge of the measured active set, and its staging slots and the measured transfer window limit how many experts it can stage. This replay is not an end-to-end DynaByte result.

At a 25% high-precision share, 12 staging slots leave 96 low-precision residents per layer and reduce exposed transfer to 0.44 ms/request. At a 40% high-precision share, exposed transfer is minimized at eight slots with 43 low-precision residents, then rises because additional staging displaces residents. Raising the high-precision share from 25% to 40% therefore changes both residency and the staging depth that reduces exposed transfer. Host-to-device bandwidth stays at the measured rate of this trace.

2.6 Motivation and Design Challenges

The four measurements are the inputs of Eq. (2). Quantization, caching, and prefetching can coexist in one system. A fixed quota or a fixed priority cannot compare the opportunity cost of fidelity, residency, and lookahead as the workload changes.

Challenge 1: heterogeneous value. The three uses are measured in different units and last for different amounts of time. Admission checks $\widehat{R}(X) \le \widehat{\epsilon}$, and measured ΔQ is reported against ϵ. $\widehat{R}$ is not converted into milliseconds or added to the latency objective.

Challenge 2: future-dependent value at different time scales. Routing locality and deadlines change faster than the quality evidence required to justify a fidelity transition. The same allocation still assigns expert memory and transfer bandwidth, while the evidence and copy cost of fidelity take longer to amortize.

Challenge 3: hard memory safety during transitions. A final resident set that fits in B can still pass through a transition that does not. Promotion and prefetch require destination bytes before

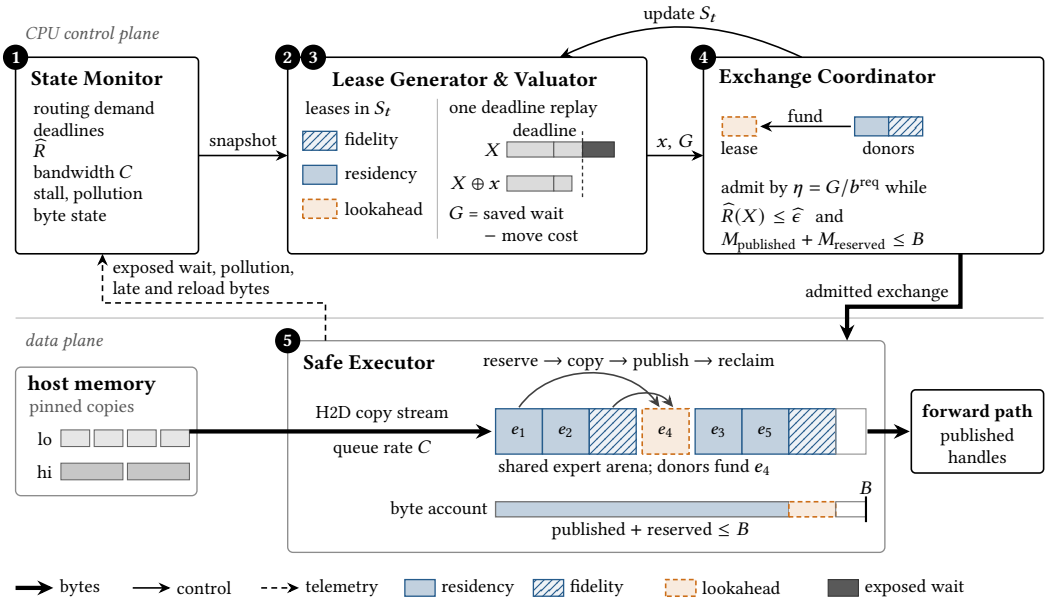


Fig. 5. DynaByte architecture. The State Monitor builds one snapshot (step 1); the Lease Generator emits fidelity, residency, and lookahead leases (step 2); the valuator scores a closed exchange with one deadline replay (step 3); and the Exchange Coordinator admits it when $\widehat{R} \le \widehat{\epsilon}$ (step 4). The Safe Executor holds the byte peak within B and commits the schedule on one host-to-device queue and one expert arena (step 5).

a copy begins, and an old published representation remains occupied until its readers finish. A representation still in transfer cannot execute.

3 DESIGN

3.1 Overview

DynaByte answers the allocation question with one admission procedure, shown in Figure 5. The procedure chooses each expert's representation, residency, and transfer issue time, then commits those choices as one exchange.

The State Monitor builds one snapshot of routing demand, deadlines, $\widehat{R}$, bandwidth, and byte state (step 1). The Lease Generator emits fidelity, residency, and lookahead leases inside the envelope S_t (step 2). The coordinator attaches donor leases, and the valuator replays the migration queue before and after the resulting exchange (step 3). The coordinator admits the exchange when predicted exposed wait falls and $\widehat{R} \le \widehat{\epsilon}$ (step 4). The executor holds the byte peak within B. The Safe Executor reserves, copies, publishes, and reclaims according to the admitted schedule (step 5). The return path updates the next snapshot and S_t.

The object passed from step 4 to step 5 is

$$x = (Z^+, D, \mathcal{T}), \tag{1}$$

where Z^+ denotes the requested leases, D the donor leases released to fund them, and $\mathcal{T}$ the transition schedule. The coordinator scores the state after the whole exchange, so the value of one use includes its effect on the other two. Peak expert memory tests whether the transition fits. $\widehat{R}$ tests the admission proxy.

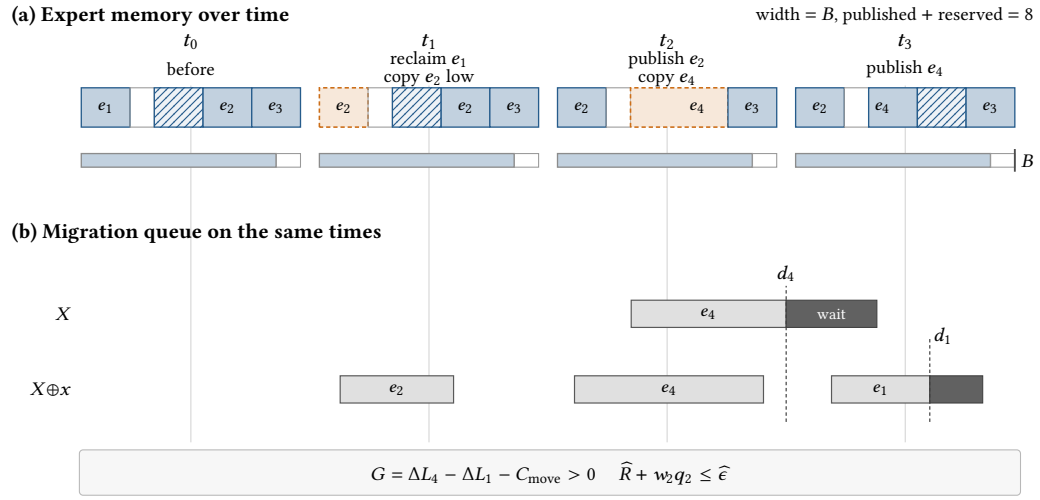


Fig. 6. A closed exchange on one timeline. (a) Every bar has the same width B. Live plus reserved bytes stay at 8 while e_1's residency and e_2's fidelity increment are released to reserve e_4. (b) On that timeline, X misses d_4, and $X \oplus x$ pays for e_2's copy and e_1's reload. The exchange is admitted because $G > 0$ and $\widehat{R} + w_2 q_2 \le \widehat{\epsilon}$.

A runtime policy π chooses the representation, resident set, and transfer issue time of every expert. The budget B is the expert memory remaining after non-expert weights, the KV cache, activations, and fixed workspaces are reserved, and C is the measured host-to-device copy rate defined in Section 2.1. The allocation problem is

$$\min_{\pi} \ \mathrm{P95}(\mathrm{TPOT}; \pi) \quad \text{subject to} \quad M_{\mathrm{peak}}(\pi) \le B, \qquad \Delta Q(\pi) \le \epsilon. \tag{2}$$

DynaByte treats Eq. (2) as that problem. The executor enforces $M_{\mathrm{peak}}(\pi) \le B$ on every transition. The controller admits with $\widehat{R}(X) \le \widehat{\epsilon}$ in place of the $\Delta Q(\pi) \le \epsilon$ test. The deadline replay charges C for the migration-queue time occupied by each exchange. The runtime consults a calibrated risk estimate while making decisions; Section 6.2 explains why this estimate is not a per-request quality guarantee.

An expert is absent, resident in the low-precision representation, resident in the high-precision representation, or in transition toward a resident state. Its packed footprints are m_i^{lo} and m_i^{hi}. A resident high-precision representation is the packed low-precision representation plus the fidelity increment $\delta m_i^{\mathrm{fid}}$ in Table 1, and it remains one packed block. Lookahead copies an absent expert before its deadline and becomes residency at the selected tier once the copy is published. The low- and high-precision representations of one absent expert are mutually exclusive. Section 3.5 specifies when a reserved representation may be published and when donor bytes may be reclaimed.

Figure 6 traces one exchange through both budgets. Lookahead e_4 does not fit in the one free unit, and e_1's residency still leaves it short, so the coordinator also releases e_2's fidelity increment. The replay admits the exchange only when e_4's saved wait exceeds the cost of e_1's later reload and e_2's low-precision copy, while the added $\widehat{R}$ remains within $\widehat{\epsilon}$.

3.2 State Monitor

The State Monitor performs step 1 in Figure 5: it turns observed runtime facts into one immutable snapshot. It neither predicts future uses nor constructs or admits an exchange. The snapshot contains demand and timing, quality, and execution and byte state.

Demand and timing. The monitor records recent routing transitions, per-layer expert-kernel time, and pre-gating logits when the checkpoint exposes them. Section 3.3 converts these observations into future-use probabilities, deadlines, and reuse estimates.

Quality. An offline sensitivity $q_i \geq 0$ is the increase in calibration loss when only expert i uses its low representation. The runtime does not modify q_i. After each forward step, the monitor updates w_i with an exponential average of expert e's router-weighted dispatch frequency. The product $w_i q_i$ estimates how often the current workload exercises that sensitivity. For a planned state X, $h_i(X) = 1$ when expert i's next execution uses the high representation. The monitor provides

$$\widehat{R}(X) = \sum_i w_i q_i \big(1 - h_i(X)\big). \tag{3}$$

The coordinator admits against $\widehat{R}(X) \leq \widehat{\epsilon}$. It does not convert risk into milliseconds.

Execution and byte state. The monitor separates the published state used by dispatch from the reserved bytes of an accepted transition, and reports their locations and byte counts separately. It also records measured bandwidth, exposed wait, late bytes, unused lookahead bytes, and eviction-induced reload bytes. Stall s_t is the fraction of routed-layer time spent waiting for expert transfers. Pollution p_t is unused lookahead bytes plus reload bytes, divided by the lookahead traffic admitted in that period. The monitor reports the exponentially smoothed values $\widetilde{s}_t$ and $\widetilde{p}_t$. Section 3.4 uses them to change S_t; the monitor does not.

The resulting snapshot is the only interface from the monitor to the Lease Generator.

3.3 Lease Generator and Valuator

The Lease Generator performs step 2 in Figure 5: it turns the monitor snapshot into leases. The Valuator performs step 3, but only after the coordinator has attached donors and thereby closed a lease into an exchange. Neither component admits an exchange.

Bound the candidate set. The envelope S_t is the maximum routed-layer depth from which the generator may produce leases (Table 1). It also reserves no lookahead quota. Section 3.4 adjusts S_t from observed stall and pollution. For a resident set R and predicted expert set $\hat{U}(S)$, the generator retains a prefix depth only when

$$\sum_{i \in \hat{U}(S) \setminus R} m_i \leq S T_\ell C. \tag{4}$$

This comparison filters candidates. The subsequent replay still uses individual issue times and deadlines.

Forecast uses and form leases. For each retained layer, the forecaster reads pre-gating logits from the snapshot when they are available and otherwise uses recent routing transitions. It produces a predicted use $j = (i_j, b_j, \omega_j, a_j, d_j)$: expert, representation tier, occurrence probability, earliest issue time, and deadline. The deadline is the latest ready time that introduces no transfer wait.

A lease is the record in Table 1. The interval guides valuation: an early use may shorten a lookahead lease, whereas a late copy retains its bytes until completion or cancellation. Table 2 shows how all three leases enter the byte account, the quality account $\widehat{R}$, and migration-queue replay.

A fidelity lease claims δm_i^{fid} beyond the base residency lease and lasts for T_{slow} forward steps. A demotion releases that increment only after the low representation is published. A residency lease renews m_i^{lo} for T_{fast} routed layers; eviction releases it after outstanding readers finish. For an absent expert, the generator emits mutually exclusive low- and high-tier lookahead leases when

Table 2. How the three uses modify the same byte account, quality account $\widehat{R}$, and migration-queue replay. Fidelity is refreshed every T_{slow} forward steps. Residency is refreshed every T_{fast} routed layers. "Queue effect" is the quantity changed in the replay of Eq. (6).

Use	Bytes	Lifetime	Risk effect	Queue effect
Fidelity	δm_i^{fid}	T_{slow}	removes $w_i q_i$	transfer size
Residency	m_i^{lo}	T_{fast}	none by itself	transfer exists
Lookahead	$m_i^{b_i}$	issue to use	selected tier	issue time

their resulting $\widehat{R}$ is feasible. A lookahead lease lasts from issue until use, after which it becomes residency at the selected tier or is reclaimed.

Value a closed exchange. The generator sends leases, not scores, to the coordinator. For each lease, the coordinator proposes one or more donor sets D and calls the valuator on the resulting exchange $x = (Z^+, D, \mathcal{T})$. The valuator replays transfers in earliest-deadline-first order. A resident expert is ready at the start of the window. An absent expert's completion time includes bytes already queued and its earliest issue time. Uses of the same expert and tier share one readiness event under continuous batching. The predicted exposed wait is

$$\widehat{L}(X) = \sum_j \omega_j \, [r_j(X) - d_j]_+. \tag{5}$$

This scheduling surrogate is calibrated against measured tail latency. It is not a direct estimate of the P95 quantile in Eq. (2).

The valuator returns

$$G(x \mid X) = \widehat{L}(X) - \widehat{L}(X \oplus x) - C_{\text{move}}(x), \tag{6}$$

where $X \oplus x$ applies the lease, donors, and transition schedule, and C_{move} is exposed transition cost. One replay prevents a retained expert and its lookahead from both receiving credit for the same avoided transfer, and it charges every released donor for its later reload.

3.4 Exchange Coordinator

The Exchange Coordinator attaches donors before valuation and makes the only admission decision. One control event has three stages: close a lease, value and admit the exchange, and refresh the next candidate set.

Close. The coordinator initializes provisional state X from the monitor snapshot. Slack, $B - M_{\text{published}} - M_{\text{reserved}}$, funds a lease directly when it is sufficient. Otherwise D is built from fidelity leases eligible for demotion, residency leases eligible for eviction, and lookahead leases eligible for cancellation. Reclaimable leases come first: a copy that has not started or is no longer useful, and a resident whose predicted reuse lies beyond S_t. A resident that is still needed can become a donor only when the replay charges its later reload. Donor construction scans leases in increasing counterfactual loss, the extra predicted wait and move cost caused by releasing the lease. In Figure 6, lookahead e_4 is closed with e_1's residency lease and e_2's fidelity increment.

A closed exchange is feasible only when the lease and its donors do not conflict, its resulting $\widehat{R}$ remains within $\widehat{\epsilon}$, and its complete transition satisfies

$$M_{\text{published}}(t) + M_{\text{reserved}}(t) \le B \qquad \forall t. \tag{7}$$

A release enters the byte account only at its scheduled reclaim time. Two exchanges therefore cannot spend the same donor, and future bytes cannot be spent as current slack.

If $\widehat{R}$ of the published state exceeds $\widehat{\epsilon}$, the coordinator first restores feasibility through promotion exchanges with the smallest replayed latency cost per unit of $\widehat{R}$ removed. This repair uses the same close and valuation steps. Once the state is feasible, a demotion can fund another lease only when the complete resulting exchange remains feasible.

Value and admit. Let b_x^{req} be the requested lease's bytes over its next lifetime, including bytes renewed by a residency lease. The coordinator ranks feasible exchanges with positive G by

$$\eta(x \mid X) = G(x \mid X)/b_x^{\mathrm{req}}. \tag{8}$$

After each admission it updates both provisional accounts, invalidates conflicting requests, and asks the valuator to replay the affected exchanges. Retaining an expert can remove its lookahead lease, and changing a representation can alter several completion times. The search stops when no feasible exchange has positive G. Eq. (2) is the offline problem. Static-Joint freezes one feasible assignment chosen on calibration statistics. Each control event admits the feasible exchange with the largest $\eta = G/b_x^{\mathrm{req}}$, where G is the drop in predicted exposed wait after the replay of the closed exchange. Because G depends on exchanges already admitted, and Eq. (2) is a tail over requests, η is a snapshot heuristic. Admission still requires $\widehat{R} \le \widehat{\epsilon}$.

Refresh. Every T_{fast} routed layers, the generator refreshes residency and lookahead leases. Every T_{slow} forward steps, it adds fidelity leases to the same set. Minimum fidelity tenure and replacement hysteresis suppress changes too short to repay their copies. At a fast event, the coordinator increases S_t by one layer when $\widetilde{s}_t - \lambda\widetilde{p}_t > \theta_h$, decreases it by one layer when the difference falls below $-\theta_h$, and leaves it unchanged otherwise. This changes only which future experts the generator may propose. Closure, replay, and η still determine admission.

The coordinator's only output is this set of admitted schedules.

3.5 Safe Executor

The Safe Executor performs step 5 in Figure 5. Its input is an admitted schedule $(Z^+, D, \mathcal{T})$. Its output is a new published state and telemetry for the next snapshot. It does not choose among fidelity, residency, and lookahead.

Reserve. The executor reserves every extent in $\mathcal{T}$ in one transaction and checks the transition peak in Eq. (7). A destination that must exist before its donor becomes reclaimable is bound to an actual extent during this transaction. If no compatible extent can be bound, or the peak exceeds B, the executor discards the schedule and leaves the published state unchanged.

Copy and publish. Copies run on the migration stream. A demotion keeps serving the high representation until the low replacement is published. An eviction-only donor is reclaimed after its readers finish. The executor atomically publishes the new handle only after the destination copy completes. Until then the forward path continues to use the old representation.

Reclaim and ready work. The executor reclaims an old extent only after its reader count reaches zero and its recorded uses complete. After routing, groups with published handles issue before groups waiting on a transfer. A missed lookahead waits only in its assigned group, and routing and top-k do not change. A speculative exchange may displace a protected resident only when the replay of its closed exchange has already charged that resident's reload.

Telemetry. The executor returns the measurements defined in Section 3.2. They update the next snapshot and do not roll back a published representation.

Algorithm 1 One DynaByte control event.

Require: snapshot, published and reserved leases, envelope S_t

if $\widetilde{s}_t - \lambda \widetilde{p}_t > \theta_h$ **then**
  $S_t \leftarrow \min(S_t + 1, S_{\max})$
else if $\widetilde{s}_t - \lambda \widetilde{p}_t < -\theta_h$ **then**
  $S_t \leftarrow \max(S_t - 1, S_{\min})$
end if
$Z \leftarrow$ residency and lookahead leases with depth $\le S_t$
if this event is a slow boundary **then**
  add fidelity leases to Z
end if
if $\widehat{R}(X) > \widehat{\epsilon}$ **then**
  close promotions of minimum replayed cost per unit of $\widehat{R}$
  admit them until $\widehat{R}(X) \le \widehat{\epsilon}$
end if
for all lease $z \in Z$ **do**
  attach donors D in increasing counterfactual loss
  close $x = (z, D, \mathcal{T})$
  if x violates $\widehat{R}$, Eq. (7), or deadlines **then**
    discard x
  else
    compute $G(x \mid X)$ and $\eta(x \mid X)$ by deadline replay
  end if
end for
while some feasible x has $G(x \mid X) > 0$ **do**
  admit $x^\star \leftarrow \arg\max_x \eta(x \mid X)$
  apply $x^\star$ to provisional X and both accounts
  drop conflicting exchanges and replay the rest
end while
submit the admitted schedules $(Z^+, D, \mathcal{T})$

4 IMPLEMENTATION

We implement DynaByte in Python with PyTorch and HuggingFace Transformers [32, 45]. Three mechanisms below are the ones Section 3 leaves to the implementation: the inputs of the forecaster, the bytes stored for each representation tier, and the placement of one extent in the shared arena.

Forecaster. For each layer retained by S_t, the forecaster in Section 3.3 reads pre-gating logits from the snapshot when the snapshot contains them. Otherwise it ranks that layer's experts by recent activation counts and returns a prefix of configured width. Low confidence lengthens this prefix and leaves S_t unchanged. When the forecast is late for the layer that is about to run, the generator uses recent routing history and the forward path continues. The admission rule reads the forecast from the steps above.

Envelope and representations. At startup S_t is the number of layer intervals required to copy one layer of absent-expert bytes at the copy rate C, clipped to the configured bounds. Section 3.4 then updates that depth from observed stall and pollution. Before a timed run, both tiers of each layer are packed into pinned host memory, and that layer's source weights are released once both packs

exist. The checkpoint fixes the tier pair, so the two representations stay the same as B changes. While a layer runs, each tier that the layer contains is issued as one grouped GEMM.

Shared arena. Each device-resident expert representation occupies one extent, a contiguous byte range that holds that representation. Layer, tier, and lease purpose share this arena. The arena is backed by one or more preallocated segments, each a contiguous buffer, and every extent is taken from a single segment. The allocator rounds a request up to a multiple of 256 bytes and selects the smallest free range, among all segments, that can hold the rounded size. When a range is returned, adjacent free ranges in the same segment are merged.

5 EVALUATION

Four questions follow: tail latency at fixed expert-memory budgets (RQ1), a shared online assignment against a fixed order and a frozen one (RQ2), which of the three removals in Figure 9 moves the Table 5 gap (RQ3), and whether the online surrogates and the controller are cheap enough to run (RQ4).

5.1 Experimental Setup

Platform and models. We evaluate on a single NVIDIA RTX A6000 GPU with 48 GB of device memory. Table 3 summarizes each checkpoint's routing structure and the two packed tiers of routed experts. For Qwen3-Next, the shared expert is not routed and remains resident with the non-expert weights.

For Qwen3-30B, the expert arena occupies 29.74 GB, corresponding to $\Phi = 1.931$. The low- and high-tier expert packs occupy 15.40 GB and 57.98 GB, respectively, giving $\rho = 3.76$; keeping both packs resident would require 73.38 GB and therefore exceeds device capacity. Host-to-device bandwidth in the replay is 24.86 GB/s.

Table 5 and the later latency figures for this model use $B = 20.0$ GB ($\Phi = 1.3$), a reserve 9.74 GB larger than the one that leaves 29.74 GB. Table 6 uses $B = 10.78$ GB ($\Phi = 0.7$).

Table 3. Configuration of the evaluated MoE models. High and low tiers are the packed representations of each routed expert.

	Qwen3-30B	Qwen3-Next-80B	Phi-3.5-MoE
Total parameters	30.5B	80B	42B
Activated / token	3.3B	3B	6.6B
High tier	FP16	INT4	FP16
Low tier	INT4	INT2	INT4
Expert weight fraction	95%	96%	96%
Layers	48	48	32
Routed experts / layer	128	512 (+1 shared)	16
Top-K	8	10	2

Workloads and quality. Latency experiments use a fixed request order drawn from a mixed workload comprising MMLU-Pro [44], GPQA [36], AIME25 [54], GSM8K [6], and HumanEval [3]. We construct workload slices according to the input/output token-length ratio: prefill-heavy requests have the largest input-to-output ratios, decode-heavy requests have the largest output-to-input ratios, and balanced requests minimize the difference between input and output lengths. Unless otherwise specified, inference uses a batch size of 8. We measure model-path time to first token

(TTFT) and time per output token (TPOT). Request queueing, tokenization, transport, serialization, offered load, and goodput are outside the measured region.

Calibration prompts determine the sensitivity estimates, predictor width, controller constants, and baseline configurations. Evaluation prompts are disjoint, and no parameter is refit on the test set. Static-Joint keeps the assignment frozen on the calibration mix. DynaByte may change the assignment on that stream.

The all-high reference executes every routed expert using the model's high-tier representation under the same runtime as DynaByte. Experts not currently resident in HBM are streamed through the same residency path used by the evaluated policies. Thus, all-high defines a fidelity reference rather than an unconstrained placement policy.

The scores in Table 4 use 200 MMLU-Pro items, 198 GPQA items, 30 AIME25 items, 200 GSM8K items, and 164 HumanEval items. Avg. weights the five accuracies by these counts.

Baselines. We organize the baselines by the resource-management decision they isolate. *Uniform-low* places every routed expert at the low tier. *Static-mixed* keeps that low-tier pack resident and spends the slack $B - P_{\text{lo}}$ on fidelity increments for the largest-q_i experts that fit, then freezes the assignment. At the Table 5 budgets $\Phi \geq 1$, lookahead is off and the measured segment transfers no expert. *Residency-only* leaves the slack idle and *Lookahead-only* stages no expert; both are the Uniform-low point. At the Table 6 budget the low-tier pack does not fit, so Static-mixed adds no fidelity increment and absent experts transfer at low precision. Seq-FR and Seq-RF test whether independently optimizing the same decisions in a fixed order is sufficient: Seq-FR assigns fidelity before residency and lookahead, whereas Seq-RF assigns residency before fidelity and lookahead. Static-Joint jointly assigns B among fidelity, residency, and lookahead on calibration statistics, keeps the feasible assignment with the lowest latency, and freezes that assignment on the evaluation stream. *Fixed-partition* instead reserves independent quotas for the three uses, testing the value of allowing memory to move across them.

For controlled comparisons, every deployable policy receives the same expert-memory budget B, expert representations, kernels, request stream, and transition interface. Baselines that require calibration receive the same calibration traces and search budget as DynaByte. HOBBIT is the complete published system [43], including mixed-precision loading, adaptive prefetching, the multi-level cache, and the low-precision miss path.

Metrics and statistics. We report time to first token (TTFT), time per output token (TPOT), and exposed expert-transfer wait. Latency runs use five warm-up iterations followed by 1,000 measured requests on each of five seeds. Policies that share a seed see the same request order. For each policy and seed, the request-level p95 TPOT is computed on that seed's 1,000 requests. The bracket beside a p95 is the 95% interval of those five seed-level values. TTFT is a point estimate, and the paper reports no interval for it. Comparisons below use the printed centers of these brackets.

The SLO column is the percentage of requests whose TPOT is at most Static-Joint's p95 TPOT on that model. A p95 line leaves the rest of that policy's own requests above it.

Replay measurements use 1,000 measured trials following two warm-up trials on a fixed WikiText routing trace and report exposed transfer wait.

Memory accounting and admission. A run is valid only if arena occupancy remains within the expert-memory budget B and its exchange log can reconstruct the reported allocation. NVML samples process-level GPU memory every 2 ms, and Table 5 reports that high-water mark as Peak.

5.2 End-to-End Quality–Latency Tradeoff

RQ1: Tail latency at fixed expert-memory budgets. We evaluate each model at a binding memory budget and compare policies along the quality–latency frontier. A configuration is considered quality-feasible when its perplexity degradation relative to All-high satisfies $\Delta\text{PPL} \leq \epsilon$. Configurations outside this bound remain visible to expose the latency–quality tradeoff, but are excluded from the feasible latency comparison.

Table 4. WikiText-2 perplexity and downstream scores. High and Low are the curve endpoints. Avg. weights the five tasks by sample size: MMLU-Pro 200, GPQA 198, AIME25 30, GSM8K 200, and HumanEval 164.

Model	Tier	PPL	MMLU-Pro	GPQA	AIME25	GSM8K	HumanEval	Avg.
Qwen3-30B	High	6.41	76.00	54.04	33.33	90.50	84.76	74.37
	Low	7.29	73.50	50.51	30.00	89.50	84.15	72.35
	Static-Joint	7.08	75.00	53.03	33.33	90.00	84.76	73.74
	DynaByte	7.03	74.50	53.03	33.33	89.50	84.76	73.49
Qwen3-Next-80B	High	5.74	76.00	72.22	70.00	87.00	85.37	79.55
	Low	6.40	71.50	65.66	60.00	86.00	84.76	76.01
	Static-Joint	6.28	75.00	71.21	66.67	87.00	85.37	78.91
	DynaByte	6.26	75.50	71.21	66.67	87.00	85.37	79.04
Phi-3.5-MoE	High	6.01	54.00	36.87	40.00	88.50	70.73	61.36
	Low	6.91	52.00	34.34	36.67	87.50	70.12	59.72
	Static-Joint	6.77	53.50	35.86	40.00	88.00	70.73	60.86
	DynaByte	6.83	53.00	35.36	40.00	88.50	70.73	60.73

Table 4 reports WikiText-2 perplexity and the five downstream tasks. DynaByte's averages are 73.49, 79.04, and 60.73, against the all-high averages 74.37, 79.55, and 61.36, lower by 0.88, 0.51, and 0.63 points (1.18%, 0.64%, and 1.03%). DynaByte is 0.25 and 0.13 points below Static-Joint on Qwen3-30B and Phi-3.5-MoE, and 0.13 points above it on Qwen3-Next-80B.

Figure 7 shows the corresponding quality–latency frontier at one binding budget per model: Qwen3-30B at $\Phi = 1.3$ ($B = 20.0$ GB), Qwen3-Next-80B at $\Phi = 1.4$ ($B = 28.0$ GB), and Phi-3.5-MoE at $\Phi = 1.2$. Table 5 uses $\epsilon = 0.70$, 0.60, and 0.85. Figure 7 draws the dashed lines at 0.7, 0.6, and 0.85.

Table 5 reports the corresponding tail-latency results. Static-mixed keeps every routed expert resident at the low tier and promotes the largest-q_i experts that fit in the slack above that pack. Lookahead is off, so the measured segment transfers no expert. Its p95 centers are 43.7 ms [43.1, 44.2], 117 ms [104, 122], and 51.9 ms [50.7, 53.4]. Uniform-low's centers are 43.8 ms [43.0, 44.6], 117 ms [104, 123], and 51.8 ms [50.3, 53.3]. Its ΔPPL values are 0.79, 0.62, and 0.86, above $\epsilon = 0.70$, 0.60, and 0.85. Those Static-mixed centers remain on Figure 7 and do not enter the quality-feasible latency comparison. On Qwen3-30B the promoted share is 13 of 128 experts per layer. On Phi-3.5-MoE the slack admits about 7% of experts. On Qwen3-Next-80B the slack admits 10% of experts. Residency-only and Lookahead-only are the Uniform-low point on these budgets. Uniform-low remains above ϵ on every model. Static-Joint freezes the calibration assignment, so the gap to DynaByte is the value of recomputing that assignment on the evaluation stream. On every model, DynaByte's p95 center is below Static-Joint's. The centers differ by 2.8 ms, 27 ms, and 5.2 ms on Qwen3-30B, Qwen3-Next-80B, and Phi-3.5-MoE. These centers are not matched in ΔPPL. On Qwen3-30B and Qwen3-Next-80B, DynaByte's ΔPPL is 0.62 and 0.52, below Static-Joint's 0.67 and 0.54, so the 2.8 ms and 27 ms gaps are not produced by spending a larger quality budget. On Phi-3.5-MoE the values are 0.82 and 0.76, both under $\epsilon = 0.85$. The 5.2 ms gap is the difference

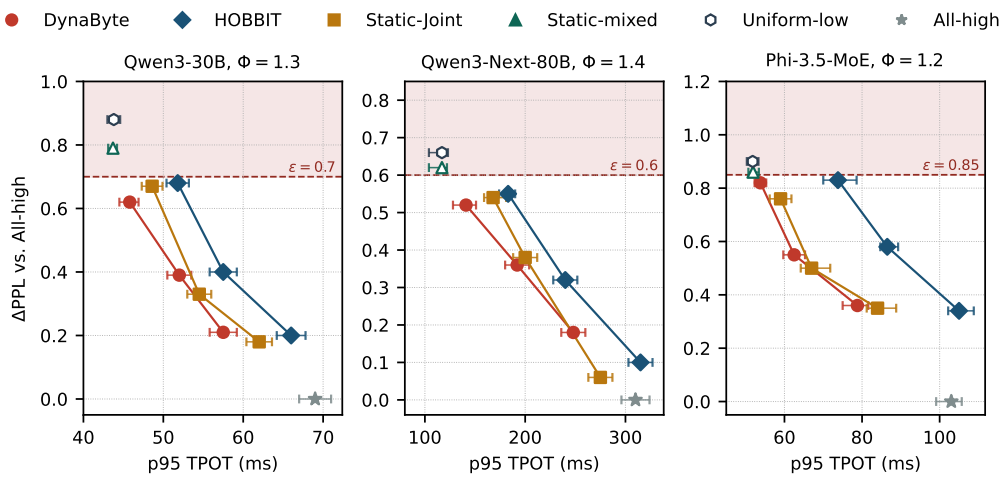


Fig. 7. Quality–latency frontier at one expert-memory budget per model. The vertical axis is ΔPPL relative to All-high, and the horizontal axis is request-level p95 TPOT. Horizontal intervals are the 95% intervals defined in Section 5.1. Table 5 lists the brackets of the selected points. Dashed lines are at 0.7, 0.6, and 0.85. Each marker is one policy setting at that budget. The selected points meet those caps; they are not a matched-ΔPPL comparison. On Phi-3.5-MoE the selected points are 0.82 and 0.76. Static-mixed keeps the low-tier pack resident and spends only the slack on fidelity; on every model its ΔPPL lies above that line.

between those two points. Satisfying the same ϵ is not the same ΔPPL. HOBBIT's p95 center is above DynaByte's on every model: 51.8 ms and 45.8 ms, 183 ms and 141 ms, and 73.8 ms and 53.8 ms.

Table 6 is the same comparison at $\Phi = 0.7$ ($B = 10.78$ GB), where 15.40 GB of low-tier experts do not fit. Every ΔPPL is 0.88, above $\epsilon = 0.70$, so none of these rows enters the quality-feasible comparison. The p95 centers are 47.0 ms for DynaByte, 53.6 ms for HOBBIT, 50.9 ms for Static-Joint, 47.9 ms for Static-mixed, and 48.4 ms for Uniform-low. The 2.8 ms, 27 ms, and 5.2 ms gaps remain the Table 5 comparisons.

5.3 Shared Online Allocation and Simpler Policies

RQ2: When does a shared online allocation improve on a simpler policy? Figure 8(a) compares DynaByte with Seq-FR and Seq-RF on a stationary balanced-length slice. Figure 8(b) records DynaByte's shares of B from the balanced slice into the decode-heavy slice. Figure 8(c) compares p95 TPOT with Static-Joint as the host-to-device cap changes.

On the balanced-length slice the sequential policies assign fidelity, residency, and lookahead in a fixed order. After the slice becomes decode-heavy, fidelity gives up bytes to lookahead.

Copy rate. The differences of the point estimates are 7.2 ms at 5 GB/s, 4.2 ms at 12 GB/s, 2.8 ms at the marked 25 GB/s point, and 0.4 ms at 40 GB/s. The marked point is the Qwen3-30B pair in Table 5: DynaByte at 45.8 ms [44.5, 46.9] and Static-Joint at 48.6 ms [47.3, 49.9]. The 5, 12, and 40 GB/s caps are shown without an interval.

5.4 Mechanism Attribution

RQ3: Which of these removals moves the Table 5 gap? We isolate the contribution of DynaByte's key mechanisms by removing one at a time while holding the workload, expert-memory budget B, and execution path fixed within each experiment. *Independent-value* makes allocation decisions using

Table 5. Selected points in Figure 7. The quality lines are $\epsilon = 0.70$, 0.60, and 0.85. Brackets are the 95% intervals defined in Section 5.1 and apply only to p95 TPOT. TTFT is a point estimate. Peak is the NVML process high-water mark. It includes non-expert weights, the KV cache, and activations, and is not the expert-memory budget B. HOBBIT and Static-Joint list the DynaByte sample at the same budget. p95 TPOT is measured on the mixed stream. ΔPPL is WikiText-2 perplexity relative to All-high for that policy.

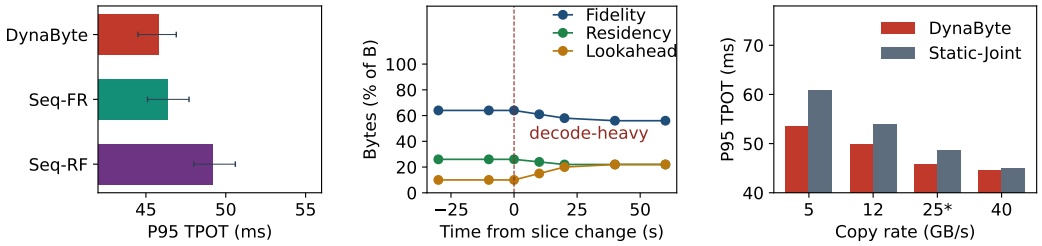
Model	Policy	ΔPPL	p95 TPOT (ms)	p95 TTFT (s)	Within SJ p95 (%)	Peak (GB)
Qwen3-30B	DynaByte	0.62	45.8 [44.5, 46.9]	3.1	97	38.6
	HOBBIT	0.68	51.8 [50.4, 53.2]	3.8	91	38.6
	Static-Joint	0.67	48.6 [47.3, 49.9]	3.2	96	38.6
	Static-mixed	0.79	43.7 [43.1, 44.2]	2.6	99	38.6
	Uniform-low	0.88	43.8 [43.0, 44.6]	2.6	99	34.0
Qwen3-Next-80B	DynaByte	0.52	141 [128, 151]	8.2	97	43.8
	HOBBIT	0.55	183 [174, 190]	10.4	94	43.8
	Static-Joint	0.54	168 [159, 181]	8.6	96	43.8
	Static-mixed	0.62	117 [104, 122]	6.8	99	43.8
	Uniform-low	0.66	117 [104, 123]	6.8	99	35.8
Phi-3.5-MoE	DynaByte	0.82	53.8 [52.3, 55.3]	2.2	96	47.1
	HOBBIT	0.83	73.8 [70.0, 78.6]	2.9	74	47.1
	Static-Joint	0.76	59.0 [56.2, 61.8]	2.3	95	47.1
	Static-mixed	0.86	51.9 [50.7, 53.4]	1.8	99	47.1
	Uniform-low	0.90	51.8 [50.3, 53.3]	1.8	99	43.8

Table 6. Qwen3-30B at $\Phi = 0.7$ ($B = 10.78$ GB), below the low-tier pack of 15.40 GB. The quality line is $\epsilon = 0.70$. Brackets are the 95% intervals defined in Section 5.1 and apply only to p95 TPOT. TTFT is a point estimate. Every row fills B, and the process high-water mark is 29.4 GB. These rows are not points of Figure 7.

Policy	ΔPPL	p95 TPOT (ms)	p95 TTFT (s)
DynaByte	0.88	47.0 [45.8, 48.2]	3.3
HOBBIT	0.88	53.6 [52.2, 55.0]	4.1
Static-Joint	0.88	50.9 [49.6, 52.2]	3.4
Static-mixed	0.88	47.9 [47.1, 48.7]	2.9
Uniform-low	0.88	48.4 [47.6, 49.2]	3.0

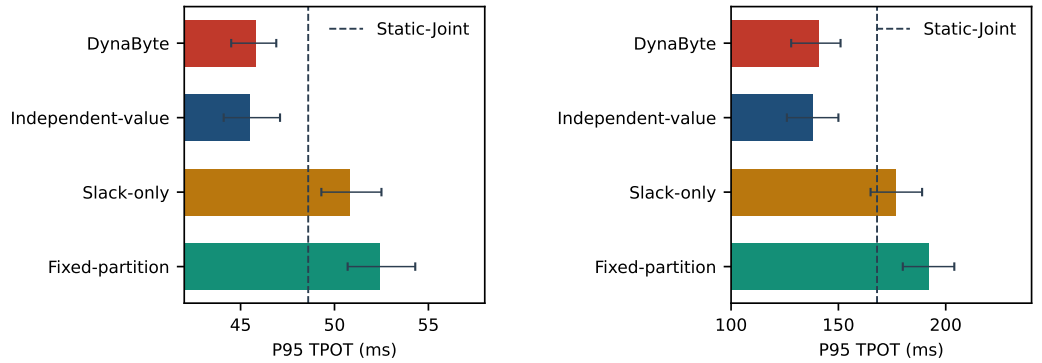
each candidate's individual value, without accounting for its effect on subsequent exchanges. *Slack-only* restricts admission to currently available slack and disables donor selection. *Fixed-partition*, defined in Section 5.1, replaces the shared arena with separate quotas for fidelity, residency, and lookahead.

On both panels of Figure 9, Slack-only's centers are 50.8 ms and 177 ms, above DynaByte's 45.8 ms and 141 ms. Fixed-partition's centers are 52.4 ms and 192 ms, above DynaByte and above the dashed Static-Joint centers. Independent-value's centers are 45.5 ms [44.1, 47.1] and 138 ms [126, 150], which are 0.3 ms and 3 ms below DynaByte's 45.8 ms [44.5, 46.9] and 141 ms [128, 151]. The 2.8 ms and 27 ms gaps are DynaByte against the Table 5 Static-Joint centers, 48.6 ms and 168 ms. On these two workloads, removing the cross-exchange term in Eq. (6) does not move the p95 center. Removing donor selection or the shared arena does.



(a) Balanced-length slice (b) Balanced to decode-heavy (c) Host-to-device copy rate

Fig. 8. When shared and online allocation is useful for Qwen3-30B at $\Phi = 1.3$. (a) Under a stationary balanced-length workload, DynaByte is compared with the two sequential orders. DynaByte's bar repeats the Table 5 point, 45.8 ms [44.5, 46.9], the 95% interval defined in Section 5.1. The Seq-FR and Seq-RF intervals use that same definition. (b) At $t = 0$, DynaByte's byte shares of B move from 64/26/10 to 56/22/22. With $\rho = 3.76$, a fidelity share f keeps a residency share of at least $f/2.76$. (c) p95 TPOT for DynaByte and Static-Joint as the host-to-device cap changes, at $\epsilon = 0.7$ and batch size 8. The marked point is the native bandwidth and repeats the Table 5 brackets, 45.8 ms [44.5, 46.9] and 48.6 ms [47.3, 49.9]. The 5, 12, and 40 GB/s bars are point estimates.



(a) Qwen3-30B, FP16/INT4 (b) Qwen3-Next-80B, INT4/INT2

Fig. 9. Ablation of DynaByte's allocation mechanisms. (a) Qwen3-30B at $\Phi = 1.3$. (b) Qwen3-Next-80B at $\Phi = 1.4$. Each ablation removes one mechanism while keeping the workload, memory budget, and execution path fixed. Horizontal bars are the 95% intervals defined in Section 5.1. Independent-value's centers are 45.5 ms [44.1, 47.1] in (a) and 138 ms [126, 150] in (b). The dashed line is the Table 5 Static-Joint center: 48.6 ms in (a) and 168 ms in (b).

5.5 Surrogate Checks and Runtime Cost

RQ4: Are the online surrogates accurate and inexpensive enough for runtime decisions? DynaByte relies on approximate signals to make allocation decisions on the critical path. Figure 10 checks predicted exposed wait, $\widehat{R}$, lookahead recall, and controller time for Qwen3-30B.

Figure 10(a) compares the latency surrogate $\widehat{L}$ with measured exposed transfer wait. The surrogate estimates the cost of an exchange from the transferred bytes and the execution window available to hide the transfer. On this workload the predictions sit above measured wait, and the panel marks a bias of +1.2 ms per token.

Figure 10(b) evaluates $\widehat{R}$ against measured ΔPPL. Of the states admitted under $\widehat{\epsilon}$, 5 of 71 have measured ΔPPL above ϵ, an unsafe-accept rate of 7.0%. Their median measured value is 1.04ϵ, and the largest is 1.12ϵ. The panel has 148 states: 66 safe accepts, 5 unsafe accepts, 18 conservative rejects, and 59 rejects. Both axes are divided by their limits. The vertical limit is $\epsilon = 0.70$. The horizontal limit is $\widehat{\epsilon}$.

Figure 10(c) measures lookahead recall at lookahead depth 3. Recall is 0.71 on prefill-heavy requests, 0.58 on mixed requests, and 0.46 on decode-heavy requests. Recall is the fraction of the subsequently demanded expert set identified early enough to influence allocation.

Finally, Figure 10(d) measures controller cost as the number of exchange candidates K and the replay window W increase. Our default operating point is $K = 8$ and $W = 32$, where replay time is 0.15 ms, below one layer. The same panel marks a band from 33 ms to 130 ms once K and W leave that point. We compare controller time with the available execution window to determine whether allocation decisions remain off the critical path, and increase K and W to characterize how the optimization cost scales beyond the default configuration.

6 DISCUSSION

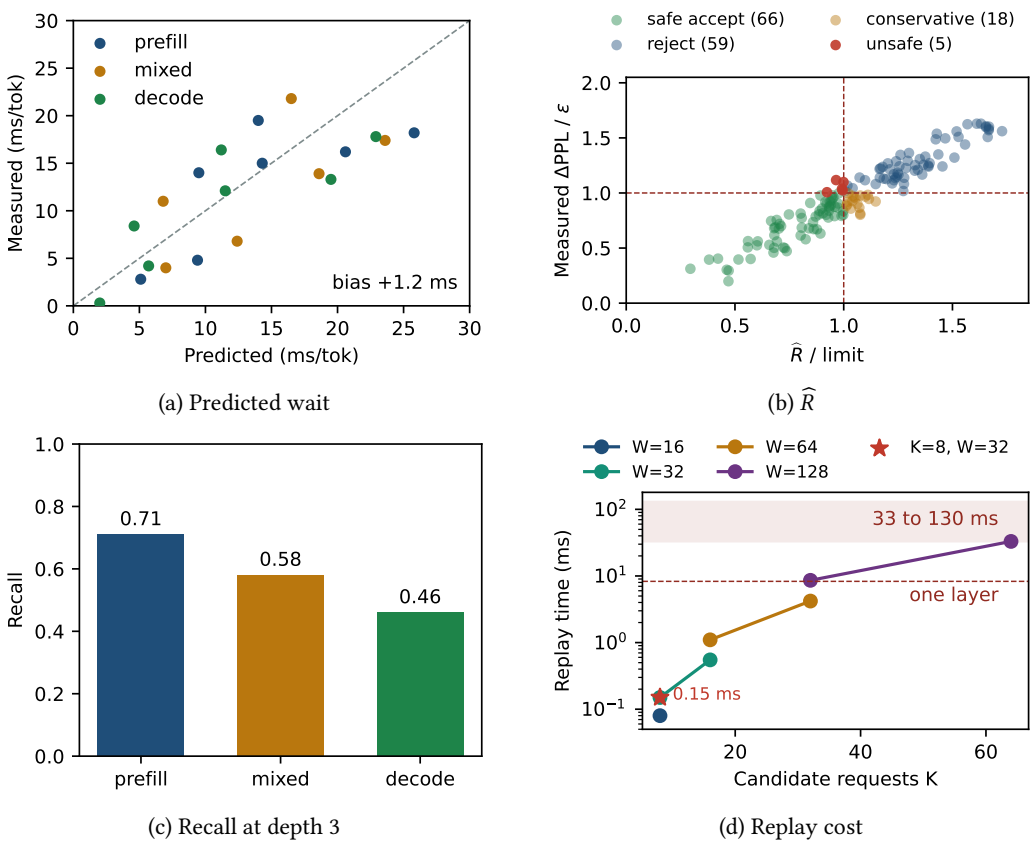
6.1 Applicability and Boundary Regimes

Joint allocation matters only when the expert-memory budget leaves a meaningful choice among fidelity, residency, and lookahead. When $\Phi \geq \rho$, every expert can remain resident in the high-precision representation and the conflict disappears. The measurements in Section 5 use budgets below that point. For Qwen3-30B, $\rho = 3.76$ and the expert arena in Section 5.1 has $\Phi = 1.931$, so a high-precision representation of every expert still does not fit. At this leftover arena the low-tier pack fits inside B, so a resident low-tier set requires no expert copy. When $1 \leq \Phi < \rho$, the aggregate low-tier footprint fits, but every fidelity increment consumes bytes that could otherwise retain or stage an expert. The three uses compete in this band. Table 5 reports that resident low-tier pack, with the slack spent on the largest q_i, as Static-mixed. Its ΔPPL is above ϵ on every model, so the quality-feasible comparison remains the one against Static-Joint. Table 5 uses the tighter budget $\Phi = 1.3$, 9.74 GB below that arena. When $\Phi < 1$, some experts must remain absent even in the low-precision representation. Table 6 is that budget for Qwen3-30B. Every listed ΔPPL is 0.88, above $\epsilon = 0.70$, so these rows do not enter the quality-feasible comparison.

Host-to-device bandwidth is the other boundary. Figure 8(c) is that boundary. Its marked native point is the Qwen3-30B row of Table 5, where the centers differ by 2.8 ms. At 40 GB/s the difference of the point estimates is 0.4 ms. Table 5 uses $\epsilon \in \{0.70, 0.60, 0.85\}$. Figure 8(c) uses $\epsilon = 0.7$. On a stationary trace, Static-Joint already places fidelity and residency on the calibration hot set. When the workload shifts from balanced to decode-heavy, Figure 8(b) shows fidelity falling and lookahead rising. Joint allocation changes that gap when copy time exceeds the overlap window. The online policy separates from Static-Joint when the routed set leaves the calibration hot set.

6.2 Guarantees and Estimation Limits

The design separates properties enforced by the executor from quantities estimated by the controller. The quality account $\widehat{R}$ combines offline sensitivity q_i, which estimates the consequence of using expert i's low-precision representation, with routing weight w_i, which estimates how often the current workload exercises that consequence. Their product is useful for admission, but it need not be additive across experts or stable across tasks. Interactions among quantized experts can invalidate a sum of individual sensitivities, and workload shift can invalidate the calibration values. The runtime admits with $\widehat{R}(X) \leq \widehat{\epsilon}$ and does not convert $\widehat{R}$ into milliseconds. The actual constraint $\Delta Q \leq \epsilon$ is evaluated on complete configurations, with $\widehat{\epsilon}$ fixed before the evaluation workload is



(a) Predicted wait

(b) $\widehat{R}$

(c) Recall at depth 3

(d) Replay cost

Fig. 10. Validation of the online decision signals and controller cost for Qwen3-30B. (a) Predicted versus measured exposed transfer wait; the diagonal denotes perfect prediction. (b) $\widehat{R}$ versus measured ΔPPL. The four markers are disjoint. Both axes are fractions of their limits, and the limits are drawn at 1. Safe accepts and unsafe accepts are the states with $\widehat{R}$ at or below $\widehat{\epsilon}$; the unsafe-accept rate divides the unsafe count by that sum. (c) Lookahead recall at lookahead depth 3. (d) Controller cost as candidate count K and replay window W increase; the star marks the default $K = 8$, $W = 32$ configuration.

observed. Figure 10(b) shows the admitted states whose measured ΔPPL exceeds ϵ. The controller applies no safety margin to $\widehat{\epsilon}$, and the executor does not roll back a published representation. Table 5 reports mixed-stream p95 and WikiText-2 ΔPPL for each listed policy at the stated budgets.

Memory safety and routing semantics do not depend on these estimates. Latency efficiency and admission under $\widehat{\epsilon}$ do. Section 5.5 shows that predicted wait sits above measured wait. A speculative exchange prices a reload only inside the current envelope, so reuse beyond that envelope can still be missed.

6.3 System Scope and Deployment Assumptions

The current runtime manages one accelerator and uses host memory as its backing store. Host memory retains both packed representations for the process lifetime; a deployment that cannot do so would need to quantize on demand or read a representation from storage, adding a transition

not modeled here. Resident-first execution assumes that expert groups within a layer can be issued independently. A kernel that issues experts in a fixed order needs a different latency estimate.

Expert-parallel execution and an SSD tier are outside the present design. They add per-device capacity, network or storage queues, and load imbalance, so the replay and the byte account would have to include those queues.

6.4 Validity, Reproducibility, and Ethics

The motivating measurements and trace replay in Sections 2.2–2.5 use configurations that differ from the joint runtime. They establish quality sensitivity, routing variation, and transfer pressure, not an end-to-end performance result for DynaByte. Claims based on their absolute values must be repeated under the final allocator, exchange policy, and memory accounting.

The used workload slices do not cover long contexts, multimodal requests, or rapidly changing request mixtures, any of which can change routing breadth, the KV reserve, and available overlap. Table 4 weights tasks by item count, so AIME25 is 30 of 792 items. One AIME item moves that task by 3.33 points and the average by 0.13 points. The per-task columns are where a change of that size is read. A device name is not a cause: the policy uses the measured bandwidth and kernel time, while kernels, runtime stacks, transfer paths, and quantization support can still differ across devices.

A result is reproducible under Eq. (2) only when the logs suffice to verify B, ϵ, and the executed state transitions.

The study uses public latency traces and quality benchmarks, recruits no participants, collects no new user data, and sends no prompts to a third-party service. Released artifacts will identify source prompts without redistributing material whose license forbids it.

7 RELATED WORK

Prior systems control fidelity, residency, or transfer timing, often while holding the other decisions fixed. We group them by that control decision, then compare DynaByte directly with systems that already combine the mechanisms.

Precision assignment. GPTQ, AWQ, SmoothQuant, LLM.int8(), OmniQuant, and SqueezeLLM fix representations before serving [9, 17, 25, 31, 37, 46]. ZeroQuant and AQLM are further offline compressors for transformer weights [13, 50]. QMoE compresses MoE weights below one bit per parameter [16]. MoE-specific methods use expert sensitivity, activation frequency, or router statistics to assign precision under a memory or quality target [4, 5, 12, 26, 53]. Such methods improve fidelity for a chosen footprint but commonly evaluate the assignment under fixed placement and transfer policies. DynaByte can use their packed representations and offline sensitivities; its additional question is whether a fidelity increment is worth the residency or lookahead displaced at runtime. Mixed-precision kernels such as MxMoE remain complementary [12]: they determine the compute cost of a selected representation, not which state should occupy expert memory.

Residency and placement. Expert caches retain recently or frequently used experts and select victims using routing history, layer position, or co-activation [2, 27, 30, 47, 51, 56]. These caches primarily estimate residency value with a fixed expert representation. Fiddler evaluates experts that are absent from GPU memory on the CPU, and SiDA builds a data-aware plan for serving large MoE models in limited accelerator memory [11, 24]. PowerInfer keeps frequently activated neurons of sparse feed-forward layers on the GPU and evaluates the remaining neurons on the CPU [42]. Placement methods colocate co-activated experts or map them according to topology and load [7, 49]; their multi-device communication objective is outside our single-accelerator implementation. DeepSpeed-MoE and Tutel run expert-parallel training and inference across

devices [21, 34]. MegaBlocks supplies grouped sparse kernels for MoE training [18]. In DynaByte, a residency lease is valued together with the fidelity or lookahead lease that would displace it, so eviction cost belongs to the complete exchange.

Prefetch and mixed-precision offloading. Cross-layer prefetchers predict future experts and overlap transfers with current computation, using either fixed lookahead or adaptation from observed demand and system state [14, 22, 41]. ExpertFlow coordinates expert scheduling and movement under offload [39]; a separate work with the same name changes token assignment to improve activation efficiency [19]. Their principal control variable is transfer timing for a given expert size and quality policy.

HOBBIT combines dynamic mixed-precision loading, adaptive prefetching, and a multi-level cache; its low-precision miss path reduces transfer cost for experts judged less critical [43]. DyMoE combines online expert importance, depth-aware mixed precision, caching, and lookahead prefetching, and some configurations omit selected experts [20]. These systems establish that the three mechanisms can coexist. The distinction is whether one exchange decides both which bytes occupy expert memory and which transfers occupy the migration queue. Measured quality is reported separately from $\widehat{\epsilon}$. Section 5.1 compares against the complete HOBBIT system under the same budget, representations, kernels, and request stream; DyMoE is not included, because the controller does not implement its estimator, depth-dependent bit schedule, or next-layer predictor.

Serving and memory management. Orca, vLLM, and SGLang schedule requests and manage KV-cache memory [28, 52, 55]. Sarathi-Serve chunks long prefills so decode iterations continue, and DistServe and Splitwise place prefill and decode on separate resources [1, 33, 57]. ZeRO-Infinity offloads training state across GPU, CPU, and NVMe, and FlexGen schedules dense-model inference across GPU, CPU, and disk [35, 40]. DynaByte operates on the expert-memory budget left after non-expert state is reserved and is complementary to those serving functions. Because a serving engine can outgrow its declared KV or activation reserve, we report the byte account and process-level peak separately.

8 CONCLUSION

DynaByte allocates expert memory and host-to-device bandwidth among fidelity, residency, and lookahead on one byte account and publishes a representation only after its copy completes. Figure 9 shows a separate quota is slower than DynaByte, and Independent-value's center stays with DynaByte. Its dashed line is the Static-Joint center in Table 5. The marked point of Figure 8(c) is that same comparison; the other caps vary host-to-device bandwidth around it and are point estimates.

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